

7. Option-Setting Memory (OFSM)

7.1 Overview

The option-setting memory (OFSM) is a collective term for the registers listed below.

- Serial programmer command control register (SPCC)
- OCD/serial programmer ID setting register (OSIS)
- Option function select register 0 (OFS0)
- Option function select register 1 (OFS1)
- Endian select register (MDE)
- TM enable flag register (TMEF)
- TM identification data register (TMINF)
- Bank select register (BANKSEL)
- Flash access window setting register (FAW)
- ROM Code Protection Register (ROMCODE)

The option-setting memory (configuration setting area) determines the state of this MCU after a reset.

The method of setting the option-setting memory is different from that of the I/O registers. For details, refer to **section 7.5, Setting the Option-Setting Memory**.

Figure 7.1 shows the option-setting memory.

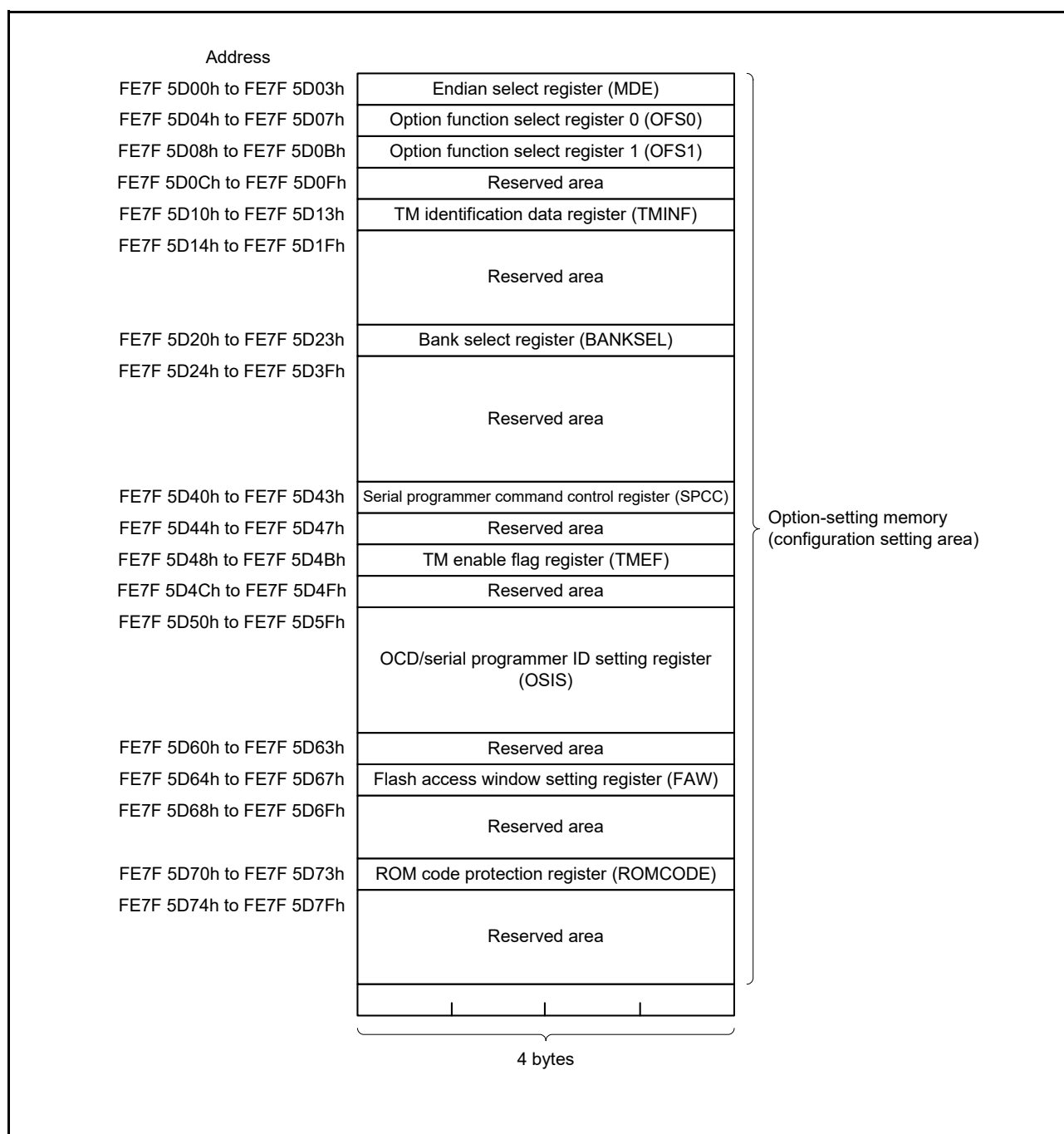


Figure 7.1 Option-Setting Memory

7.2 Register Descriptions

7.2.1 Serial Programmer Command Control Register (SPCC)

Address(es): OFSM.SPCC FE7F 5D40h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	SPE	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b26 to b0	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b27	SPE	Serial Programmer Connection Enable	0: Connection of a serial programmer is prohibited after a reset. 1: Connection of a serial programmer is permitted after a reset.	R
b31 to b28	—	Reserved	When reading, these bits return to the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

This register is used to permit or prohibit the connection of a serial programmer.

SPE Bit (Serial Programmer Connection Enable)

This bit enables or disables the connection of a serial programmer.

Resetting MCU after setting this bit to 0 disables the connection of a serial programmer.

7.2.2 OCD/Serial Programmer ID Setting Register (OSIS)

This register is used to store the control code or ID code for the on-chip debugger ID code protection and serial programmer ID code protection.

After the emulator or serial programmer sends a control code or ID code, it is tested for a match with the value stored in this register.

Connection to the emulator/serial programmer can proceed if the codes match and cannot proceed if they do not.

For the blank product, the value after a reset for ID code 1/control code to ID code 16 is FFh. These registers are set to the specified values after setting a value.

Address	Bit 31			Bit 0
FE7F 5D50h to FE7F 5D53h	ID Code 4	ID Code 3	ID Code 2	ID Code 1/Control Code
FE7F 5D54h to FE7F 5D57h	ID Code 8	ID Code 7	ID Code 6	ID Code 5
FE7F 5D58h to FE7F 5D5Bh	ID Code 12	ID Code 11	ID Code 10	ID Code 9
FE7F 5D5Ch to FE7F 5D5Fh	ID Code 16	ID Code 15	ID Code 14	ID Code 13

ID Code 1/Control Code to ID Code 16

The control code or ID code for the on-chip debugger ID code protection and serial programmer ID code protection is stored in this register.

ID code 1 is used as a control code for connection to a serial programmer and as an ID code for connection to an emulator.

For details of the control code, refer to section 7.4, Settings of the Option-Setting Memory and Read, Program, and Erase Operations.

7.2.3 Option Function Select Register 0 (OFS0)

Address(es): OFSM.OFS0 FE7F 5D04h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	WDTRS TIRQS	WDTRPSS[1:0]	WDTRPES[1:0]	WDTCKS[3:0]			WDTTOS[1:0]			WDTST RT	—		

Value after reset:

The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	IWDTSLCSTP	—	IWDTIRSTIRQS	IWDTRPSS[1:0]	IWDTRPES[1:0]	IWDTCKS[3:0]			IWDTTOS[1:0]			IWDTST TRT	—		

Value after reset:

The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	When reading, this bit returns the value written by the user. The write value should be 1.	R
b1	IWDTSTRT	IWDT Start Mode Select	0: IWDT is automatically activated in auto-start mode after a reset 1: IWDT is halted after a reset	R
b3, b2	IWDTTOS[1:0]	IWDT Timeout Period Select	b3 b2 0 0: 1024 cycles (03FFh) 0 1: 4096 cycles (0FFFh) 1 0: 8192 cycles (1FFFh) 1 1: 16384 cycles (3FFFh)	R
b7 to b4	IWDTCKS[3:0]	IWDT-Dedicated Clock Frequency Division Ratio Select	b7 b4 0 0 0 0: No division 0 0 1 0: Divide-by-16 0 0 1 1: Divide-by-32 0 1 0 0: Divide-by-64 1 1 1 1: Divide-by-128 0 1 0 1: Divide-by-256 Settings other than above are prohibited.	R
b9, b8	IWDTRPES[1:0]	IWDT Window End Position Select	b9 b8 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (No window end position setting)	R
b11, b10	IWDTRPSS[1:0]	IWDT Window Start Position Select	b11 b10 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (No window start position setting)	R
b12	IWDTIRSTIRQS	IWDT Reset Interrupt Request Select	0: Non-maskable interrupt request or plain interrupt request is enabled 1: Reset is enabled	R
b13	—	Reserved	When reading, this bit returns the value written by the user. The write value should be 1.	R
b14	IWDTSLCSTP	IWDT Sleep Mode Count Stop Control	0: Counting stop is disabled 1: Counting stop is enabled when entering sleep, software standby, deep software standby, or all-module clock stop mode	R
b16, b15	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b17	WDTSTRT	WDT Start Mode Select	0: WDT is automatically activated in auto-start mode after a reset 1: WDT is stopped after a reset	R

Bit	Symbol	Bit Name	Description	R/W
b19, b18	WDTTOPS[1:0]	WDT Timeout Period Select	b19 b18 0 0: 1024 cycles (03FFh) 0 1: 4096 cycles (0FFFh) 1 0: 8192 cycles (1FFFh) 1 1: 16384 cycles (3FFFh)	R
b23 to b20	WDTCKS[3:0]	WDT Clock Frequency Division Ratio Select	b23 b20 0 0 0 1: Divide-by-4 0 1 0 0: Divide-by-64 1 1 1 1: Divide-by-128 0 1 1 0: Divide-by-512 0 1 1 1: Divide-by-2048 1 0 0 0: Divide-by-8192 Settings other than above are prohibited.	R
b25, b24	WDRPES[1:0]	WDT Window End Position Select	b25 b24 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (No window end position setting)	R
b27, b26	WDRPSS[1:0]	WDT Window Start Position Select	b27 b26 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (No window start position setting)	R
b28	WDRSTIRQS	WDT Reset Interrupt Request Select	0: Non-maskable interrupt request or plain interrupt request is enabled 1: Reset is enabled	R
b31 to b29	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

IWDTSTRT Bit (IWDT Start Mode Select)

This bit selects the mode in which the IWDT is activated after a reset (stopped state or activated in auto-start mode). When activated in auto-start mode, the OFS0 register setting for the IWDT is effective.

IWDTTOPS[1:0] Bits (IWDT Timeout Period Select)

These bits select the timeout period, i.e. the time it takes for the down-counter to underflow, as 1024, 4096, 8192, or 16384 cycles of the frequency-divided clock set by the IWDTCKS[3:0] bits. The time (number of clock cycles for the IWDT) it takes to underflow after a refresh operation is determined by the combination of the IWDTCKS[3:0] bits and IWDTTOPS[1:0] bits.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

IWDTCKS[3:0] Bits (IWDT-Dedicated Clock Frequency Division Ratio Select)

These bits select, from 1/1, 1/16, 1/32, 1/64, 1/128, and 1/256, the division ratio of the prescaler to divide the frequency of the clock for the IWDT. Using the setting of these bits together with the IWDTTOPS[1:0] bit setting, the IWDT counting period can be set from 1024 to 4194304 clock cycles for the IWDT.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

IWDRPES[1:0] Bits (IWDT Window End Position Select)

These bits select the position of the end of the window for the down-counter as 0%, 25%, 50%, or 75% of the value being counted by the counter. The value of the window end position must be smaller than the value of the window start position (window start position > window end position). If the value for the window end position is greater than the value for the window start position, only the value for the window start position is effective.

The counter values corresponding to the settings for the start and end positions of the window in the IWDRPSS[1:0] and IWDRPES[1:0] bits vary with the setting of the IWDTTOPS[1:0] bits.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

IWDRPSS[1:0] Bits (IWDT Window Start Position Select)

These bits select the position where the window for the down-counter starts as 25%, 50%, 75%, or 100% of the value being counted (the point at which counting starts is 100% and the point at which an underflow occurs is 0%). The interval between the positions where the window starts and ends becomes the period in which refreshing is possible, and refreshing is not possible outside this period.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

IWDRSTIRQS Bit (IWDT Reset Interrupt Request Select)

The setting of this bit selects the operation on an underflow of the down-counter or generation of a refresh error. An independent watchdog timer reset, a non-maskable interrupt request, or an interrupt request is selectable.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

IWDTSLCSTP Bit (IWDT Sleep Mode Count Stop Control)

This bit selects to stop counting when entering sleep, software standby, deep software standby, or all-module clock stop mode.

For details, refer to section 35, Independent Watchdog Timer (IWDTa).

WDTSTRT Bit (WDT Start Mode Select)

This bit selects the mode in which the WDT is activated after a reset (stopped state or activated in auto-start mode).

When activated in auto-start mode, the OFS0 register setting for the WDT is effective.

WDTTOPS[1:0] Bits (WDT Timeout Period Select)

These bits select the timeout period, i.e. the time it takes for the down-counter to underflow, as 1024, 4096, 8192, or 16384 cycles of the frequency-divided clock set by the WDTCKS[3:0] bits. The time (number of PCLKB cycles) it takes to underflow after a refresh operation is determined by a combination of the WDTCKS[3:0] bits and WDTTOPS[1:0] bits.

For details, refer to section 34, Watchdog Timer (WDTA).

WDTCKS[3:0] Bits (WDT Clock Frequency Division Ratio Select)

These bits select, from 1/4, 1/64, 1/128, 1/512, 1/2048, and 1/8192, the division ratio of the prescaler to divide the frequency of PCLKB. Using the setting of these bits together with the WDTTOPS[1:0] bit setting, the WDT counting period can be set from 4096 to 134217728 PCLKB cycles.

For details, refer to section 34, Watchdog Timer (WDTA).

WDRPES[1:0] Bits (WDT Window End Position Select)

These bits select the position of the end of the window on the down-counter as 0%, 25%, 50%, or 75% of the value being counted by the counter. The value of the window end position must be smaller than the value of the window start position (window start position > window end position). If the value for the window end position is greater than the value for the window start position, only the value for the window start position is effective.

The counter values corresponding to the settings for the start and end positions of the window in the WDRPSS[1:0] and WDRPES[1:0] bits vary with the setting of the WDTTOPS[1:0] bits.

For details, refer to section 34, Watchdog Timer (WDTA).

WDRPSS[1:0] Bits (WDT Window Start Position Select)

These bits select the position where the window for the down-counter starts as 25%, 50%, 75%, or 100% of the value being counted (the point at which counting starts is 100% and the point at which an underflow occurs is 0%). The

interval between the positions where the window starts and ends becomes the period in which refreshing is possible, and refreshing is not possible outside this period.

For details, refer to [section 34, Watchdog Timer \(WDTA\)](#).

WDRSTIRQS Bit (WDT Reset Interrupt Request Select)

The setting of this bit selects the operation on an underflow of the down-counter or generation of a refresh error. A watchdog timer reset, a non-maskable interrupt request, or an interrupt request is selectable.

For details, refer to [section 34, Watchdog Timer \(WDTA\)](#).

7.2.4 Option Function Select Register 1 (OFS1)

Address(es): OFSM.OFS1 FE7F 5D08h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	HOCO EN	—	—	—	—	—	LVDAS	VDSEL[1:0]	—

Value after reset: The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b1, b0	VDSEL[1:0]	Voltage Detection 0 Level Select	b1 b0 0 0: Reserved 0 1: Selects 2.94 V 1 0: Selects 2.87 V 1 1: Selects 2.80 V	R
b2	LVDAS	Voltage Detection 0 Circuit Start	0: Voltage monitor 0 reset is enabled after a reset 1: Voltage monitor 0 reset is disabled after a reset	R
b7 to b3	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b8	HOCOEN	HOCO Oscillation Enable	0: HOCO oscillation is enabled after a reset 1: HOCO oscillation is disabled after a reset	R
b31 to b9	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

VDSEL[1:0] Bits (Voltage Detection 0 Level Select)

These bits select the voltage detection level of the voltage detection 0 circuit.

LVDAS Bit (Voltage Detection 0 Circuit Start)

This bit selects whether the voltage monitor 0 reset is enabled or disabled after a reset.

HOCOEN Bit (HOCO Oscillation Enable)

This bit selects whether the HOCO oscillation enable bit is effective or not after a reset.

Setting the HOCOEN bit to 0 allows the HOCO oscillation to be started before the CPU starts operation, and therefore reduces the waiting time for oscillation stabilization.

Note that even if the HOCOEN bit is set to 0, the system clock source is not switched to HOCO. The system clock source is switched to HOCO only by modifying the clock source select bits (SCKCR3.CKSEL[2:0]) from the CPU.

7.2.5 Endian Select Register (MDE)

Address(es): OFSM.MDE FE7F 5D00h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	BANKMD[2:0]			—	MDE[2:0]		

Value after reset: The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	MDE[2:0]	Endian Select	b2 b0 0 0 0: Big endian 1 1 1: Little endian Settings other than above are prohibited.	R
b3	—	Reserved	When reading, this bit returns the value written by the user. The write value should be 1.	R
b6 to b4	BANKMD[2:0]	Bank Mode Select	b6 b4 0 0 0: Dual mode 1 1 1: Linear mode Settings other than above are prohibited.	R
b31 to b7	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

This register selects the endian for the CPU and bank mode of the dual bank function of the code flash memory.

MDE[2:0] Bits (Endian Select)

These bits select little endian or big endian for the CPU.

BANKMD[2:0] Bits (Bank Mode Select)

These bits select bank mode of the dual bank function of the code flash memory. These bits cannot be rewritten while the TM function is enabled. Set the value of these bits while the TM function is disabled.

7.2.6 TM Enable Flag Register (TMEF)

Address(es): OFSM.TMEF FE7F 5D48h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	TMEFDB[2:0]			—	TMEF[2:0]			—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b23 to b0	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b26 to b24	TMEF[2:0]	TM Enable	b26 b24 0 0 0: The TM function for blocks 8 and 9 in the code flash memory is enabled. 1 1 1: The TM function for blocks 8 and 9 in the code flash memory is disabled. Settings other than above are prohibited.	R
b27	—	Reserved	When reading, this bit returns the value written by the user. The write value should be 1.	R
b30 to b28	TMEFDB[2:0]	Dual-Bank TM Enable	b30 b28 0 0 0: The TM function for blocks 78 and 79 in the code flash memory is enabled in dual mode. 1 1 1: The TM function for blocks 78 and 79 in the code flash memory is disabled in dual mode. Settings other than above are prohibited.	R
b31	—	Reserved	When reading, this bit returns the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

This register is used to enable the TM function in the code flash memory.

To enable the TM function, refer to section 62.15.23, Configuration Program Command. When the TMEF[2:0] bits or TMEFDB[2:0] bits are rewritten with the TM function enabled, rewriting the bits is ignored.

To disable the TM function, refer to section 62.15.21, Configuration Clear Command.

TMEF[2:0] Bits (TM Enable)

These bits enable or disable the TM function in the code flash memory. When enabling the TMEF[2:0] bits (000b) in dual mode, also enable the TMEFDB[2:0] (000b).

TMEFDB[2:0] Bits (Dual-Bank TM Enable)

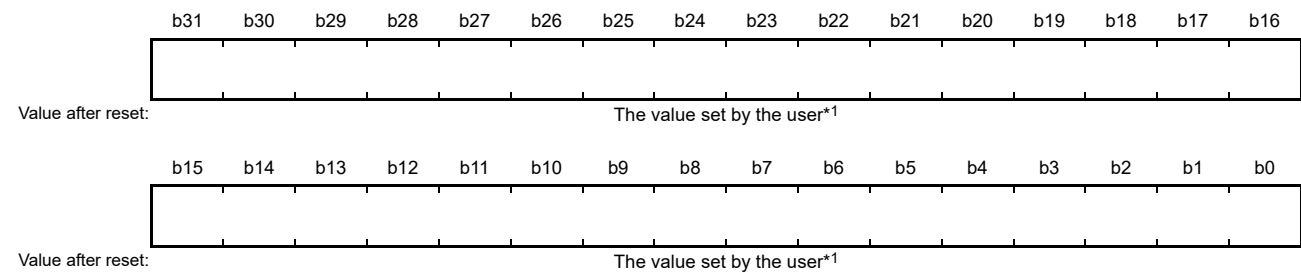
These bits enable or disable the TM function of the code flash memory in dual mode. Set these bits to disable dual mode (111b) if linear mode is to be used.

When enabling the TMEFDB[2:0] bits, also enable the TMEF[2:0] bits.

When the TMEF[2:0] bits are 111b (with the TM function disabled), this setting is disabled.

7.2.7 TM Identification Data Register (TMINF)

Address(es): OFSM.TMINF FE7F 5D10h



Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

The user can store any desired 32-bit value in this register.
This register is used to store codes that identify the program stored in the TM-target area.
When the TMINF register is rewritten by serial programming while the TM function is enabled, rewriting this register is ignored. To erase the contents of the TMINF register, refer to section 62.15.21, Configuration Clear Command.

7.2.8 Bank Select Register (BANKSEL)

Address(es): OFSM.BANKSEL FE7F 5D20h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: The value set by the user*1

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	BANKSWP[2:0]		

Value after reset: The value set by the user*1

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	BANKSWP[2:0]	Startup Bank Switch	b2 b0 0 0: These bits specify the address range of bank 1 from FFE0 0000h to FFFF FFFFh and bank 0 from FFC0 0000h to FFDF FFFFh. 1 1: These bits specify the address range of bank 1 from FFC0 0000h to FFDF FFFFh and bank 0 from FFE0 0000h to FFFF FFFFh. Settings other than above are prohibited.	R
b31 to b3	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

This register selects the startup bank of the program when the code flash memory is in dual mode.

BANKSWP[2:0] Bits (Startup Bank Switch)

These bits select the address range of the bank 0 and bank 1 of the code flash memory when the code flash memory is in dual mode.

Selecting the addresses of bank 0 and bank 1 selects the program to be started up.

The setting of these bits in linear mode is invalid.

For details of the startup bank selection, refer to section 62.10.5.2, Selecting the Startup Bank.

7.2.9 Flash Access Window Setting Register (FAW)

Address(es): OFSM.FAW FE7F 5D64h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
BTFLG	—	—	—	FAWE[11:0]											
Value after reset: The value set by the user*1															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
FSPR	—	—	—	FAWS[11:0]											
Value after reset: The value set by the user*1															

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	FAWS[11:0]	Flash Access Window Start Address*2	Flash access window start address	R
b14 to b12	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b15	FSPR	Access Window Protection	0: With protection 1: Without protection	R
b27 to b16	FAWE[11:0]	Flash Access Window End Address*2	Flash access window end address	R
b30 to b28	—	Reserved	When reading, these bits return the value written by the user. The write value should be 1.	R
b31	BTFLG	Start-up Area Select	0: FFFF C000h to FFFF DFFFh are used as the start-up area 1: FFFF E000h to FFFF FFFFh are used as the start-up area	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

Note 2. Set the value of the FAWS[11:0] bits to be no greater than that of the FAWE[11:0] bits.

If the value of the FAWS[11:0] bits is equal to that of the FAWE[11:0] bits, the entire user area of the code flash memory becomes programmable and erasable.

This register is used to set the write protection bit and start-up area select bit for setting the flash access window start address, flash access window end address, and access window.

FAWS[11:0] Bits (Flash Access Window Start Address)

These bits are used to set the access window start address.

The access window can be set in block units.

Set the FAWS[10:0] bits to b23 to b13 of the access window start address. Set the FAWS[11] bit to 0.

FSPR Bit (Access Window Protection)

Setting the FSPR bit protects the following operations.

- Setting the areas including the FAW register by using the configuration setting command of the FSCI commands.
- Setting the areas including the FAW register by using the configuration program command in boot mode
- Erasing the option-setting memory by using the configuration clearing command in boot mode
- Changing the setting of the start-up area protection by using the FSUACR register.
- Erasing the all data in the flash memory when results of judgment did not match 3 times in a row while the control code is set to 45h in boot mode.

Once 0 is written to this bit, the bit can never be restored to 1.

Therefore, the access window and the start-up area protection never be set again or the TM function never be disabled once it has been enabled.

Exercise extra caution when handling the FSPR bit.

FAWE[11:0] Bits (Flash Access Window End Address)

These bits are used to set the access window end address.

The access window can be set in block units.

Set the FAWE[10:0] bits to b23 to b13 of the access window end address. Set the FAWE[11] bit to 0. However, if the access window end address is 1 0000 0000h (= FFFF FFFFh + 1), set the FAWE[11:0] bits to 800h.

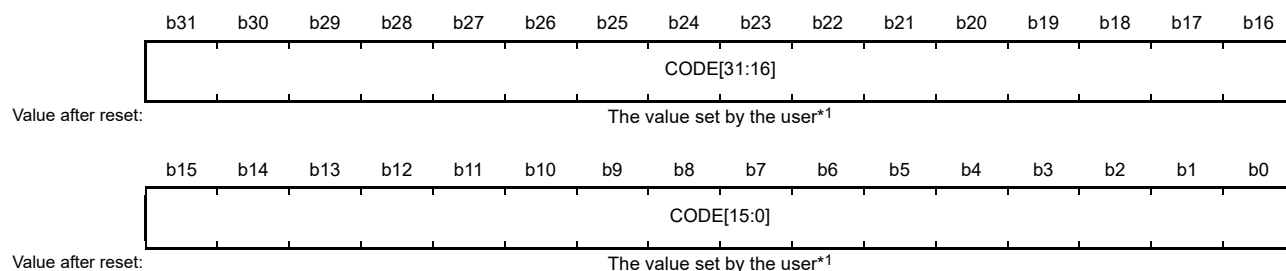
BTFLG Bit (Start-up Area Select)

This bit is used to set whether the start-up area is switched by using the start-up program protection. In dual mode (the MDE.BANKMD[2:0] bits are 000b), write 1 to this bit.

For details, refer to section 62.10.3, Start-Up Program Protection.

7.2.10 ROM Code Protection Register (ROMCODE)

Address(es): OFSM.ROMCODE FE7F 5D70h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	CODE[31:0]	ROM Code	0000 0000h: ROM code protection enabled (ROM code protection 1) 0000 0001h: ROM code protection enabled (ROM code protection 2) Other than above: ROM code protection disabled	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after setting a value.

This register is used to disable read, program, and erase operations of the flash memory by a parallel programmer being used for off-board programming.

The ROM code in flash memory is a 32-bit code.

Table 7.1 shows the specifications for ROM code protection.

For release from ROM code protection, write FFFF FFFFh (code protection is disabled) to the ROM code by using the configuration setting command of the self-programming, or by using the configuration setting command of the boot mode, or erase the ROM code by using the configuration clearing command. The user can store any desired 32-bit value in the register.

Table 7.1 Specifications for ROM Code Protection

ROM Code	State of Protection	Operations at the Time of Connection with the Parallel Programmer
0000 0000h	ROM code protection enabled (ROM code protection 1)	Reading, programming, and erasing of the code flash memory is prohibited.
0000 0001h	ROM code protection enabled (ROM code protection 2)	Reading from the code flash memory is prohibited.
Other than above	ROM code protection disabled	Reading, programming, and erasing of the code flash memory is permitted.

7.3 Programming and Erasing of the Option-Setting Memory in Individual Operating Modes

Table 7.2 shows programming and erasing of the option-setting memory in the individual operating modes.

Table 7.2 Programming and Erasing of the Option-Setting Memory in Individual Operating Modes

Option-Setting Memory	ROMCODE Setting Value	Boot Mode (SCI Interface, USB Interface, FINE Interface)		Self-Programming		Parallel Programmer	
		Program	Erase	Program	Erase	Program	Erase
ROM code protection register (ROMCODE)	0000 0000h	✓*1	✓*1	✓*2	x	x	x
	0000 0001h	✓*1	✓*1	✓*2	x	x*3	✓*4, *5
	Other than above	✓*1	✓*1	✓*2	x	✓*4	✓*4
SPCC register, OSIS register, MDE register, OFS0 register, OFS1 register, TMEF register, TMINF register, BANKSEL register, and FAW register	0000 0000h	✓*1	✓*1	✓*2	x	x	x
	0000 0001h	✓*1	✓*1	✓*2	x	✓*4	✓*4, *5
	Other than above	✓*1	✓*1	✓*2	x	✓*4	✓*4

✓: Possible

x: Not possible

Note 1. The commands for boot mode (SCI, USB, and FINE interfaces) are used for programming or erasing. For details, refer to section 62.11, Boot Mode.

Note 2. The configuration setting command is used for programming. For how to use the configuration setting command, refer to section 62.8.3.14, Configuration Set Command.

Note 3. The ROM code cannot be programmed when the ROM code protection is set.

Note 4. The parallel programmer is used for programming and erasing. For details, refer to the manual of the parallel programmer you are using.

Note 5. It is erasable when the code flash memory is blank.

7.4 Settings of the Option-Setting Memory and Read, Program, and Erase Operations

Table 7.3 shows the settings of the option-setting memory and read, program, and erase operations when the MCU is connected to a serial programmer.

Table 7.4 shows the settings of the option-setting memory and judgment on ID codes when the MCU is connected to an OCD.

Table 7.3 Settings of the Option-Setting Memory and Read, Program, and Erase Operations When the MCU is Connected to a Serial Programmer

No.	SPCC.SPE	OSIS (Control Code)	OSIS (ID Code 2 to 16)	Connection to a Serial Programmer	Read, Program and Erase Operations after Connecting a Serial Programmer
1	0	Any value	Any value	Connection prohibited	—
2	1	45h	Any value	Control codes or ID codes matched: Transition to the command waiting phase Control codes or ID codes unmatched: Transition to the state of waiting for a serial programming ID code check command again. However, when results of judgment do not match 3 times in a row, the all data in the flash memory will be erased*1	Permitted: read, program, erase
3	1	Other than 45h	Any value	ID codes matched: Transition to the command waiting phase ID codes unmatched: Transition to the state of waiting for a serial programming ID code check command again.	Permitted: read, program, erase

Note 1. When the FAW.FSPR bit is 0, the blocks in those areas are not erased.

Table 7.4 Settings of the Option-Setting Memory and Judgment on ID Codes When the MCU is Connected to an OCD

No.	SPCC.SPE	OSIS (ID Code 1)	OSIS (ID Code 2 to 16)	Connection to an OCD
1	—	Any value	Any value	ID codes matched: Connection to an OCD is permitted ID codes unmatched: Waiting for input of ID code

7.5 Setting the Option-Setting Memory

7.5.1 Allocation of Data in the Option-Setting Memory

Data for programming in the option-setting memory should be allocated to the addresses shown in Figure 7.1. An example of source code for setting the option-setting memory is shown below.

Note: Programming formats vary depending on the compiler. Refer to the compiler manual for details.

Setting F7FFFFFFh in the serial programmer command control register (SPCC)

```
.ORG 0FE7F5D40H  
.LWORD 0F7FFFFFFFH
```

Setting the following ID codes in the OCD/serial programmer ID setting register (OSIS)

ID code 1/control code = FFh, ID code 2 = 02h, ID code 3 = 03h, ID code 4 = 04h,
ID code 5 = 05h, ID code 6 = 06h, ID code 7 = 07h, ID code 8 = 08h,
ID code 9 = 09h, ID code 10 = 0Ah, ID code 11 = 0Bh, ID code 12 = 0Ch,
ID code 13 = 0Dh, ID code 14 = 0Eh, ID code 15 = 0Fh, ID code 16 = 10h
.ORG 0FE7F5D50H
.LWORD 040302FFH, 08070605H, 0C0B0A09H, 100F0E0DH

Setting EF67BA5Dh in the option function select register 0 (OFS0)

```
.ORG 0FE7F5D04H  
.LWORD 0EF67BA5DH
```

Setting FFFFFFFEFAh in the option function select register 1 (OFS1)

```
.ORG 0FE7F5D08H  
.LWORD 0FFFFFFEFAH
```

Setting FFFFFFFF8h in the endian select register (MDE)

```
.ORG 0FE7F5D00H  
.LWORD 0FFFFFFF8H
```

Setting F7FC77F9h in the flash access window setting register (FAW)

```
.ORG 0FE7F5D64H  
.LWORD 0F7FC77F9H
```

7.6 Usage Note

7.6.1 Data for Programming Reserved Areas and Reserved Bits in the Option-Setting Memory

When reserved areas and reserved bits in the option-setting memory are within the scope of programming, write 1 as the value for all bits of reserved areas and all reserved bits. Normal operation cannot be guaranteed if 0 is written to such bits.